

A Cubic-Root-of-Unity Structure for the Kappa Invariant

Exact N -Body Formulae, Elementary-Symmetric Reduction, and Controlled-Variable Transitions

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Abstract

We study a family of symmetric polynomial invariants arising from the ratio $[P_N = \frac{\prod_{i=1}^N (1+x_i^3)}{\prod_{i=1}^N (1+x_i)}, \quad \kappa_N = 1-P_N]$. Direct symbolic calculations for $N=3,4,5$ exhibit large polynomial expressions which can be factorized. The central observation is the factorization $[1+x^3 = (1+x)(1-x+x^2)]$, which yields the exact all- N identity $[P_N = \prod_{i=1}^N (1-x_i+x_i^2)]$. Writing $\omega = e^{2\pi i/3}$, this expression admits the root-of-unity representation $[P_N = E(\omega)E(\omega^2) = |E(\omega)|^2, \quad E(t) = \prod_{i=1}^N (1+x_i t) = \sum_{k=0}^N e_k t^k]$. Consequently, if the elementary symmetric polynomials $1 - A^2 - B^2 - C^2 + AB + AC + BC$ are known, then the exact transition law obtained by adjoining a controlled variable n : $[\kappa_{N+1} - \kappa_N = \dots]$ can be determined. In particular, adjoining 0 or 1 leaves the invariant unchanged. For a two-factor integer $N=pq$, the same structure is observed.

Contents

1	Introduction	2
2	Definitions	2
3	The Fundamental Factorization	2
4	Cubic Roots of Unity	3
5	Universal Elementary-Symmetric Formula	3
6	Recovery of the $N = 3, 4, 5$ Formulae	3
7	Controlled-Variable Transition	4
8	The Special Values $z = 0$ and $z = 1$	4
9	Multiple Controlled Variables	4
10	The Two-Factor Case and a Possible Factoring Connection	4
11	Controlled Auxiliary Factors and Factor Recovery	5
12	Why the Controlled Variables Are Interesting	5

13 Interpretation of the Elementary-Symmetric Pattern	5
14 Generalization to Higher Powers	6
15 Computational Verification	6
16 Limitations Regarding Integer Factorization	6
17 Research Program	7
17.1 All- N symbolic verification	7
17.2 Factor-recovery experiment	7
17.3 Non-circular accessibility	7
17.4 Controlled two-variable probes	7
17.5 Complexity analysis	7
18 Conclusion	7

1 Introduction

The purpose of this paper is to describe an exact algebraic structure behind a family of symmetric polynomial expressions that initially appear to grow rapidly in complexity with the number of variables.

For three variables p, q, r , define $[M_{1,3} = (1+p)(1+q)(1+r)]$ and $[X_3 = (1+p^3)(1+q^3)(1+r^3)]$. The corresponding normalized quantity is $[P_3 = \frac{X_3}{M_{1,3}}, \quad \kappa_3 = 1 - P_3]$.

Analogous definitions are made for arbitrary N : $[M_{1,N} = \prod_{i=1}^N (1+x_i)]$, $[X_N = \prod_{i=1}^N (1+x_i^3)]$, $[P_N = \frac{X_N}{M_{1,N}}, \quad \kappa_N = 1 - P_N]$.

Direct computer algebra calculations for $N = 3, 4, 5$ give exact polynomial quotients and verify that the resulting κ_N are symmetric polynomials. Their elementary-symmetric representations are surprisingly compact.

The main purpose of this paper is to explain why.

The essential identity is elementary: $[1+x^3 = (1+x)(1-x+x^2)]$. It immediately implies $[P_N = \prod_{i=1}^N (1-x_i+x_i^2)]$. The quadratic factor has a second important representation: $[1-x+x^2 = (1+\omega x)(1+\omega^2 x)]$ where $[\omega^3 = 1, \quad \omega \neq 1, \quad 1+\omega+\omega^2 = 0]$.

This leads to a root-of-unity evaluation of the elementary-symmetric generating polynomial and explains the complete quadratic structure observed experimentally.

2 Definitions

Definition 2.1 (Elementary symmetric polynomials). For variables $x_1, \dots, x_N$, define $[e_0 = 1]$ and, for $1 \leq k \leq N$, $[e_k = \sum_{1 \leq i_1 < \dots < i_k \leq N} x_{i_1} \cdots x_{i_k}]$. Thus $[\prod_{i=1}^N (1+x_i t) = \sum_{k=0}^N e_k t^k]$.

Definition 2.2 (Kappa invariant). For $N \geq 1$, define $[M_{1,N} = \prod_{i=1}^N (1+x_i)]$, $[X_N = \prod_{i=1}^N (1+x_i^3)]$ and $[P_N = \frac{X_N}{M_{1,N}}, \quad \kappa_N = 1 - P_N]$.

The original expressions written as sums over all subsets are $[M_{1,N} = \sum_{S \subseteq \{1, \dots, N\}} \prod_{i \in S} x_i]$ and $[X_N = \sum_{k=0}^N e_k(x_1^3, \dots, x_N^3)]$.

3 The Fundamental Factorization

Theorem 3.1 (Exact product representation). For arbitrary N , $[P_N = \prod_{i=1}^N (1-x_i+x_i^2)]$ and therefore $[\kappa_N = 1 - \prod_{i=1}^N (1-x_i+x_i^2)]$.

Proof. For every variable x , $[1+x^3 = (1+x)(1-x+x^2)]$. Therefore $[X_N = \prod_{i=1}^N (1+x_i^3)]$.

$[\prod_{i=1}^N (1+x_i)(1-x_i+x_i^2)]$. Dividing by $[M_{1,N} = \prod_{i=1}^N (1+x_i)]$ gives $[P_N = \prod_{i=1}^N (1-x_i+x_i^2)]$. The formula for κ_N follows from $[\kappa_N = 1 - P_N]$. $\square$

This identity shows that the large rational expressions produced by direct symbolic division conceal a simple multiplicative invariant.

4 Cubic Roots of Unity

Let $[\omega = e^{2\pi i/3}]$ Then $[\omega^3 = 1, \quad \omega \neq 1, \quad 1 + \omega + \omega^2 = 0.]$

We have $[(1+\omega x)(1+\omega^2 x) = 1 + (\omega + \omega^2)x + \omega^3 x^2 = 1 - x + x^2.]$
Hence:

Theorem 4.1 (Root-of-unity representation). *Let $[E(t) = \prod_{i=1}^N (1 + x_i t) = \sum_{k=0}^N e_k t^k]$ Then $[P_N = E(\omega)E(\omega^2)]$ and $[P_N = |E(\omega)|^2.]$*

Proof. Using $[1 - x_i + x_i^2 = (1 + \omega x_i)(1 + \omega^2 x_i)]$ we obtain $[P_N = \prod_i (1 + \omega x_i) \prod_i (1 + \omega^2 x_i)]$. The first product is $E(\omega)$ and the second is $E(\omega^2)$. For real variables, $[E(\omega^2) = \overline{E(\omega)}]$ which gives the modulus – square representation. $\square$
This is the central structural identity of the paper.

5 Universal Elementary-Symmetric Formula

Since $[E(\omega) = \sum_{k=0}^N e_k \omega^k]$ and the powers of ω are periodic modulo 3, define $[A_N = \sum_{0 \leq k \leq N \equiv 0 \pmod{3}} e_k, B_N = \sum_{0 \leq k \leq N \equiv 1 \pmod{3}} e_k, C_N = \sum_{0 \leq k \leq N \equiv 2 \pmod{3}} e_k]$.
Then $[E(\omega) = A_N + B_N \omega + C_N \omega^2.]$

Theorem 5.1 (Universal elementary-symmetric form). *For arbitrary N , $[P_N = A_N^2 + B_N^2 + C_N^2 - A_N B_N - A_N C_N - B_N C_N]$ and hence $[\kappa_N = 1 - A_N^2 - B_N^2 - C_N^2 + A_N B_N + A_N C_N + B_N C_N]$*

Proof. We have $[P_N = |A_N + B_N \omega + C_N \omega^2|^2]$. Since $[\omega + \omega^2 = -1]$ and $[|\omega|^2 = 1]$, direct multiplication gives $[P_N = A_N^2 + B_N^2 + C_N^2 - A_N B_N - A_N C_N - B_N C_N]$. Subtracting from 1 gives the expression for κ_N . $\square$

An equivalent and sometimes more revealing expression is $[P_N = \frac{1}{2}[(A_N - B_N)^2 + (B_N - C_N)^2 + (C_N - A_N)^2]]$

Thus $[\kappa_N = 1 - \frac{1}{2}[(A_N - B_N)^2 + (B_N - C_N)^2 + (C_N - A_N)^2]]$

This representation makes the modulo-3 structure explicit.

6 Recovery of the $N = 3, 4, 5$ Formulae

The symbolic calculations for three, four, and five variables are special cases of the universal theorem.

For $N = 3$, $[A_3 = e_0 + e_3 = 1 + e_3, \quad B_3 = e_1, \quad C_3 = e_2]$. Substitution gives $[\kappa_3 = -e_1^2 + e_1 e_2 + e_1 e_3 - 2e_1 e_4 + e_1 e_5]$

For $N = 4$, $[A_4 = 1 + e_3, \quad B_4 = e_1 + e_4, \quad C_4 = e_2]$. Therefore $[\kappa_4 = -e_1^2 + e_1 e_2 + e_1 e_3 - 2e_1 e_4 + e_1 e_5]$

For $N = 5$, $[A_5 = 1 + e_3, \quad B_5 = e_1 + e_4, \quad C_5 = e_2 + e_5]$. Hence $[\kappa_5 = -e_1^2 + e_1 e_2 + e_1 e_3 - 2e_1 e_4 + e_1 e_5]$

These are precisely the exact expressions obtained by direct symbolic division and elementary-symmetric reduction.

7 Controlled-Variable Transition

The most useful consequence for increasing the number of variables is the multiplicative transition law.

Theorem 7.1 (Exact one-variable transition). *Let* $\prod_{i=1}^N (1 - x_i + x_i^2)$ *If a new variable is appended, then*

$$1 - \prod_{i=1}^N (1 - x_i + x_i^2) = (1 - z + z^2)(1 - \kappa_{N+1})$$

Proof. From the product representation, $1 - \kappa_{N+1} =$

$$\frac{P_{N+1}}{P_N(1 - z + z^2)}.$$

Since $P_N = 1 - \kappa_N$, we obtain $1 - \kappa_{N+1} = \frac{(1 - z + z^2)(1 - \kappa_N)}{1 - (1 - z + z^2)}$

$$= \frac{z(1 - z)(1 - \kappa_N)}{z(1 - z)} = 1 - \kappa_N.$$

This explains the experimentally observed quadratic dependence on the new variable.

Indeed, $z(1 - z) = z - z^2$.

8 The Special Values $z = 0$ and $z = 1$

Corollary 8.1. *For every* N , $\kappa_{N+1}(x_1, \dots, x_N, 0) = \kappa_N(x_1, \dots, x_N)$

and $\kappa_{N+1}(x_1, \dots, x_N, 1) = \kappa_N(x_1, \dots, x_N)$.

Proof. For $z = 0$ or $z = 1$, $1 - z + z^2 = 1$. The result follows immediately from the transition theorem. $\square$

Thus the zeros of $z(1 - z)$ are exact fixed points of the extension operation.

9 Multiple Controlled Variables

The one-variable transition iterates immediately.

Theorem 9.1 (Exact controlled extension). *Let* $r_1, \dots, r_m$ *be appended to an* N -*variable system. Then* $1 - \kappa_{N+m} = (1 - \kappa_N) \prod_{j=1}^m (1 - r_j + r_j^2)$ *Equivalently,*

$$\kappa_{N+m} = 1 - (1 - \kappa_N) \prod_{j=1}^m (1 - r_j + r_j^2).$$

Proof. Apply the one-variable transition repeatedly: $1 - \kappa_{N+1} =$

$$\frac{(1 - r_1 + r_1^2)(1 - \kappa_N)}{1 - (1 - r_1 + r_1^2)} = \frac{(1 - r_2 + r_2^2)(1 - \kappa_{N+1})}{1 - (1 - r_2 + r_2^2)},$$

and so on. Multiplication gives the result. $\square$

In particular, for a two-body base system with variables p, q , $1 - \kappa_{p,q,r} = (1 - \kappa_{p,q})(1 - r + r^2)$

and $1 - \kappa_{p,q,r,s} = (1 - \kappa_{p,q})(1 - r + r^2)(1 - s + s^2)$.

The controlled variables therefore contribute known multiplicative factors which can, in principle, be divided out.

10 The Two-Factor Case and a Possible Factoring Connection

Suppose $N = pq$, where p and q are unknown factors of a known integer N .

Set $u = p + q$. The elementary symmetric polynomials are $e_1 = u$, $e_2 = N$.

For two variables, $P_2 = (1 - p + p^2)(1 - q + q^2)$. Direct expansion gives $P_2 = u^2 - Nu - u + N^2 - N + 1$.

Hence $\kappa_2 = -u^2 + Nu + u - N^2 + N$.

Therefore, if P_2 or κ_2 were available independently, then $u = p + q$ could be obtained from a quadratic equation.

From $[P_2 = u^2 - (N+1)u + N^2 - N + 1,]$ we have $[u^2 - (N+1)u + (N^2 - N + 1 - P_2) = 0.]$

Once u is known, p and q are the roots of $[x^2 - ux + N = 0.]$ Thus $[p, q = u \pm \sqrt{u^2 - 4N}.]$

This establishes a precise algebraic connection between the invariant and factor recovery.

11 Controlled Auxiliary Factors and Factor Recovery

Now suppose $[N=pq]$ is known but p, q are unknown, while an auxiliary variable r is under experimental or computational control.

For the three-variable system, $[1 - \kappa_3 = (1 - \kappa_2)(1 - r + r^2).]$

Therefore $[1 - \kappa_2 = \frac{1 - \kappa_3(r)}{1 - r + r^2}.]$

Similarly, for two controlled variables r, s , $[1 - \kappa_2 = \frac{1 - \kappa_4(r, s)}{(1 - r + r^2)(1 - s + s^2)}.]$

]

Consequently, if an independent mechanism can provide $\kappa_3(r)$ or $\kappa_4(r, s)$ without first determining p and q , then the controlled-variable identity gives an exact method for recovering κ_2 , hence $p + q$, and finally p, q .

This leads to the following research problem.

Conjecture 11.1 (Invariant-access factoring principle). *If there exists an efficient procedure which, given a composite integer $N = pq$ and a freely chosen auxiliary value r , computes the corresponding invariant $\kappa_3(p, q, r)$ without implicitly performing the factorization of N , then the transition identity can be used to recover $p + q$ and hence factor N efficiently.*

The conjecture is intentionally stated as an algorithmic conditional. The algebraic identities proved in this paper do *not* by themselves establish an efficient factoring algorithm.

12 Why the Controlled Variables Are Interesting

The controlled-variable construction provides a natural normalization test.

For every admissible r , $[1 - \kappa_3(r) = \frac{1 - \kappa_2}{1 - r + r^2}.]$

Thus the right-hand side is independent of r .

For two controlled variables, $[1 - \kappa_4(r, s) = \frac{1 - \kappa_2}{(1 - r + r^2)(1 - s + s^2)}.]$

Hence measurements or computations at different control values should collapse onto the same invariant.

For example, $[1 - \kappa_3(r_1) = \frac{1 - \kappa_2}{1 - r_1 + r_1^2}.]$

$[1 - \kappa_3(r_2) = \frac{1 - \kappa_2}{1 - r_2 + r_2^2}.]$

Likewise, $[1 - \kappa_4(r_1, s_1) = \frac{1 - \kappa_2}{(1 - r_1 + r_1^2)(1 - s_1 + s_1^2)}.]$

$[1 - \kappa_4(r_2, s_2) = \frac{1 - \kappa_2}{(1 - r_2 + r_2^2)(1 - s_2 + s_2^2)}.]$

This makes controlled auxiliary variables useful as probes of the same underlying two-body quantity.

13 Interpretation of the Elementary-Symmetric Pattern

The original symbolic results show apparently complicated coefficients such as $[e_1 e_2, e_1 e_3, -2e_1 e_4, e_1 e_5,]$ and

The root-of-unity representation explains these coefficients.

Since $[P_N = \sum_{i,j} e_i e_j \omega^{i-j}]$

the coefficient associated with a pair (i,j) depends only on $[i - j \pmod{3}]$.

Therefore the coefficient pattern is not accidental and does not require separate derivation for every N .

The three residue classes $[k \equiv 0, 1, 2 \pmod{3}]$ *are the fundamental structural objects.*

This observation replaces the apparent complexity of the expanded polynomials by a three-class decomposition.

14 Generalization to Higher Powers

The cubic case is part of a broader root-of-unity construction.

For a positive integer m , $[1+x^m]$ *can be related to roots of unity and corresponding cyclotomic factors. More generally, $\prod_i (1+x_i t)$*

The present paper focuses on $m = 3$, because $[1-x+x^2]$ *is a quadratic factor and the corresponding root-of-unity structure produces an especially simple quadratic invariant.*

A natural direction for future work is to determine which other cyclotomic quotients produce similarly low-dimensional symmetric invariants.

15 Computational Verification

The symbolic calculations for $N = 3, 4, 5$ provide exact verification of the general theorem.

The following properties were verified algebraically:

1. Exact cancellation of $X_N/M_{1,N}$ for $N = 3, 4, 5$.
2. Exact construction of $\kappa_N = 1 - P_N$.
3. Exact elementary-symmetric reduction.
4. Zero symmetrization remainder.
5. Exact equality between direct and symmetric representations.
6. Exact $3 \rightarrow 4$ transition.
7. Exact $4 \rightarrow 5$ transition.
8. Quadratic dependence on the new variable.
9. Vanishing of the transition at $n = 0$ and $n = 1$.

The symbolic results are therefore consistent with the all- N factorization theorem proved above.

Importantly, however, the all- N theorem itself does not require computer algebra. It follows directly from $[1+x^3 = (1+x)(1-x+x^2)]$ *and the cubic-root-of-unity factorization.*

16 Limitations Regarding Integer Factorization

It is important to distinguish an exact invariant from an efficient factorization algorithm.

For a known composite integer $[N=pq]$, the value $[\kappa_2]$ *contains information about $p+q$. If κ_2 can be obtained without knowing p and q , factor recovery follows through a quadratic equation.*

However, if calculating κ_2 from its defining expression requires knowledge of p and q , then no computational advantage has been obtained.

Likewise, adding controlled variables r, s does not by itself solve the problem. The identities
 $[1 - \kappa_3 = (1 - \kappa_2)(1 - r + r^2)]$ and $[1 - \kappa_4 = (1 - \kappa_2)(1 - r + r^2)(1 - s + s^2)]$ only become algorithmically significant if the corresponding higher-body invariant is independently accessible.

Thus the open problem is not the algebraic identity. The identity is proved. The open problem is whether there exists a useful, non-circular way to access the invariant for a composite integer.

17 Research Program

The results suggest the following sequence of investigations.

17.1 All- N symbolic verification

Implement the formula $[\kappa_N = 1 - A_N^2 - B_N^2 - C_N^2 + A_N B_N + A_N C_N + B_N C_N]$ and verify it automatically for larger N without expanding the original expression.

17.2 Factor-recovery experiment

For selected composite integers $[N = pq]$, calculate the invariant using several controlled values r and verify the exact normalization $[1 - \kappa_3(r) = \frac{1 - \kappa_2}{1 - r + r^2}]$.

17.3 Non-circular accessibility

The central computational question should then be:

Can the quantity $\kappa_3(N, r)$ be computed from N and r without first determining p and q ?

If the answer is no for the direct polynomial definition, investigate whether an alternative representation, oracle, recurrence, modular evaluation, or physical interpretation supplies the required information.

17.4 Controlled two-variable probes

Study $[\kappa_4(N, r, s)]$ and determine whether varying (r, s) produces information that is not already available from the invariant.

17.5 Complexity analysis

If an accessible form is found, compare its computational complexity with classical factoring algorithms. In particular, determine whether the invariant merely repackages the same difficulty as factoring or actually provides a new efficient route.

18 Conclusion

The calculations for three, four, and five variables reveal an exact all-dimensional structure.

The defining quotient $[P_N = \kappa_N]$

$$\frac{\prod_i (1 + x_i^3)}{\prod_i (1 + x_i)} \text{ reduces identically to } \boxed{P_N = \prod_i (1 - x_i + x_i^2).}$$

Using a nontrivial cubic root of unity ω , $[1 - x + x^2 = (1 + \omega x)(1 + \omega^2 x)]$, so that $\boxed{P_N = E(\omega)E(\omega^2)}$ where $E(t) = \sum_{k=0}^N e_k t^k$.

Grouping the elementary symmetric polynomials by index modulo 3 gives the universal formula $\boxed{\kappa_N = 1 - A_N^2 - B_N^2 - C_N^2 + A_N B_N + A_N C_N + B_N C_N.}$

The transition under adjoining a controlled variable z is exactly $\boxed{\kappa_{N+1} - \kappa_N =}$

For several controlled variables, $\boxed{1 - \kappa_{N+m} = (1 - \kappa_N) \prod_{j=1}^m (1 - r_j + r_j^2).}$

Thus the apparent complexity of the $3 \rightarrow 4 \rightarrow 5$ transitions is resolved by a simple multiplicative law.

For the two-factor problem $N = pq$, the invariant is directly related to $p + q$, so access to the invariant could in principle lead to factor recovery. The controlled-variable identities show precisely how auxiliary variables can be divided out.

The remaining question is algorithmic rather than algebraic:

$\boxed{\text{Can the relevant invariant be accessed efficiently without already knowing the hidden factors?}}$

Answering that question is the natural next stage of this research.

Status of the results. The algebraic identities and all- N formulae presented here are exact theorems. The proposed connection to integer factorization is a research direction and does not constitute, by itself, a proven polynomial-time factoring algorithm.